LeepTalent Level 3 End-point Assessment

Learner Guide

Loan Request Data — what to do, why, and where you are in the process |
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This guide shows one sound way to complete the scenario, the reasoning behind each step, and — importantly — **which stage of the data analysis process** each step belongs to. Assessors look for evidence that you followed the process in order; learners who jump straight to answers without working through the stages often lose marks. Work the task yourself first, then check your method here. Screenshots are illustrative mock-ups; Excel is shown, but Power Query or Google Sheets reach the same outcome.

The data analysis process — follow it in order:

Identify → Collect → Clean/Prepare → Analyse → Visualise → Interpret/Report

Each step below is tagged with the stage it sits in, so you can see the process being followed and evidence it to the assessor. Don't skip stages, and don't merge them — work through them in sequence.

Scenario 1 — Data Gathering

Step 1 — Identify what you need

STAGE 1 · IDENTIFY

What you do: Before touching the tool, confirm what the task requires: three sources (`Loan Data.csv` , `Personal Data.xlsx` , the Annex A Branch Reference table), the common fields that link them (**Loan_ID**; **Property_Area**), and the end goal (one aggregated set named **'Loan Request Data'**).

> **Why start here?** Identifying the sources, keys and outcome first means your import and merge are deliberate, not trial-and-error. Assessors credit candidates who show they understood the requirement before manipulating data.

Step 2 — Bring each source in as a connection

STAGE 2 · COLLECT

What you do: Import each source as a **connection only** — don't load any of them to a worksheet yet. For each file:

1. Go to `Data` tab → `Get Data` → `From File` → `From Text/CSV` (CSV) or `From Excel Workbook` (.xlsx).
2. Select the file and click `Import` .

3. At the load prompt, choose `Only Create Connection`.
4. Repeat for all three sources (Loan Data, Personal Data, Branch Reference).

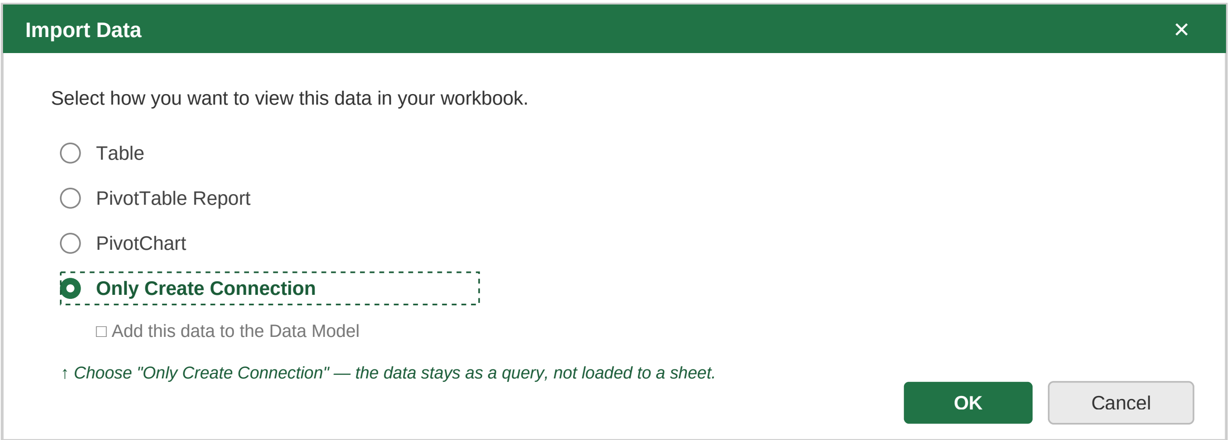

*Figure 1 — Choose **Only Create Connection** so the source stays as a query rather than being written to a sheet.*

> **Why connection only, not open the file?** Opening a CSV gives a disconnected copy — no query, no refresh, no record of method. Loading every file to its own sheet clutters the workbook with raw data you don't need. **Connection only** keeps each source as a managed, refreshable query in the background, ready to be cleaned and merged — which is what evidences the Collect stage properly.

Step 3 — Double-click to open, then clean

STAGE 3 · CLEAN / PREPARE

What you do: In the **Queries & Connections** pane, **double-click** a query to open it in Power Query. There, clean and prepare each source:

1. Check and set each column's **data type** (text, whole number, currency, date).
2. Format **Applicant Income** and **Loan Amount** as currency.
3. Remove every record with one or more blank fields (*Remove Empty / Remove Rows*).

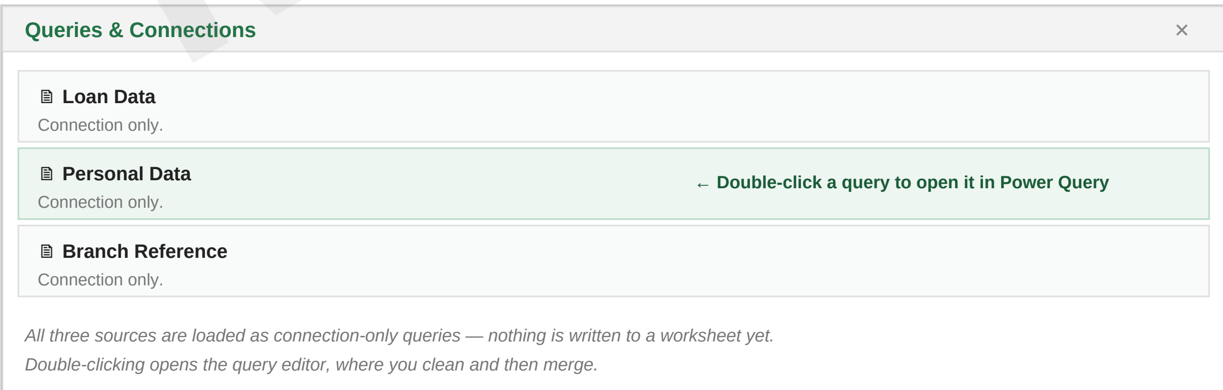

Figure 2 — All three sources as connection-only queries. Double-click a query to open it in Power Query.

> **Why clean here, before merging?** Incomplete records distort counts, totals and charts — a record missing Gender can't be counted by gender. Cleaning each query first means the merge combines trustworthy data. Doing it out of order (merging or analysing dirty data) is a common reason candidates' figures come out wrong.

> **Expected result:** the raw set of **614** records reduces to **517** complete records once all rows containing blanks are removed.

Step 4 — Merge, then Close & Load

STAGE 3 · CLEAN / PREPARE

What you do: Still in Power Query, **merge** the cleaned queries. Bring the applicant columns from Personal Data into Loan Data by matching on **Loan_ID**, and the branch columns from Branch Reference by matching on **Property_Area**. Name the combined query **'Loan Request Data'**, then `Close & Load` it to a table.

> **Why match on a key, and why merge last?** A unique key (Loan_ID) guarantees each applicant's details land on the right record — pasting columns side by side misaligns the moment the files are sorted differently. Merging only after each source is clean means one correct, complete table is loaded to the sheet, ready for analysis.

Scenario 2 — Data Analysis and Validation

Step 5 — Derive the Monthly Payment

STAGE 4 · ANALYSE

What you do: Insert a column to the right of **Loan Amount Term** headed **Monthly Payment**, and enter a formula dividing the loan amount by its term. Format to Currency, 2 dp.

```
=[@[Loan Amount]] / [@[Loan_Amount_Term]]
```

E2 fx `=[@[Loan Amount]] / [@[Loan_Amount_Term]]`

Loan Amount	Loan_Amount_Term	Monthly Payment	Credit_History
128	36	£3.56	1
66	36	£1.83	1
120	36	£3.33	1

The result is a live formula, not a typed number — it recalculates if the data changes.

Figure 3 — Monthly Payment as a live formula.

> **Why a formula, not a typed value?** A formula recalculates if the data changes and proves the result was derived, not guessed. Typed numbers break the moment a loan amount is corrected and give the marker no evidence of method.

Step 6 — Count the applicants

STAGE 4 · ANALYSE

What you do: In **O2** and **O3**, count each gender with COUNTIF:

`=COUNTIF(Table1[Gender],"Male")` and `=COUNTIF(Table1[Gender],"Female")`

> **Why COUNTIF?** It counts straight from the live column, so totals update if records change — far more reliable than counting by eye on 500+ rows.

Expected result (clean set): Male = **424**, Female = **93** (517 total).

Step 7 — Visualise the finding

STAGE 5 · VISUALISE

What you do: Summarise total Loan Amount for married males vs married females (a PivotTable filtered to **Married = Yes**, or two SUMIFS), then insert a bar or column chart.

`=SUMIFS(Table1[Loan Amount],Table1[Gender],"Male",Table1[Married],"Yes")`

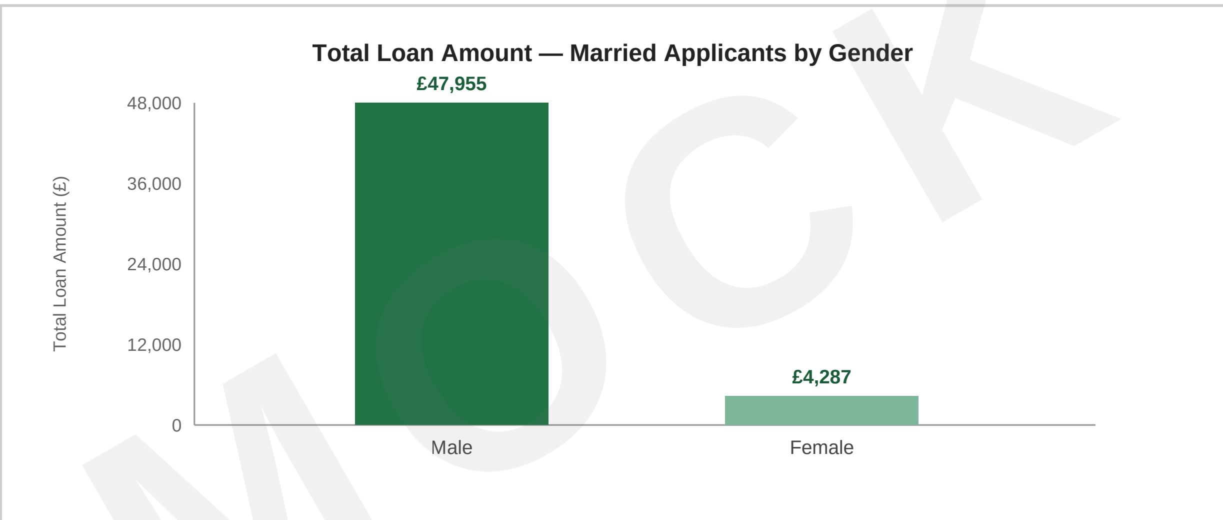

Figure 4 — Total loan amount for married applicants, split by gender.

> **Why visualise?** A chart makes the imbalance obvious instantly — senior managers read a bar chart in seconds where a table of numbers takes longer. The visual carries the finding.

Married applicants	Total Loan Amount	Count
Male	£47,955.00	306
Female	£4,287.00	27

Loan amounts are in the data's own units (thousands £); present them consistently with how the company reports loan values.

Step 8 — Interpret and report

STAGE 6 · INTERPRET / REPORT

What you do: Write a short findings note beneath the chart for senior management, then save as an Excel workbook (keeping the table, formulas and chart intact).

> **Why this stage matters most for marks:** data only has value once someone can act on it. A plain-English interpretation turns numbers into an insight a non-technical reader can use. Saving as .xlsx (not CSV) preserves the live formulas, table and chart — exporting to CSV here would strip all of them.

"After cleaning, 517 complete loan requests remain. Male applicants account for the large majority (424 of 517, ~82%). Among married applicants, total lending to males (£47,955) is roughly eleven times that to females (£4,287), driven mainly by volume — 306 married male requests against 27 married female. This points to a heavily male-weighted applicant base that the business may wish to investigate."

Marker's checklist — process followed in order

- **Identify:** sources, keys and goal confirmed before manipulating data.
- **Collect:** data brought in as **connection only** (not opened); double-clicked to open, cleaned, then merged on correct keys.
- **Clean/Prepare:** currency formatting applied; blank records removed (614 → 517).
- **Analyse:** Monthly Payment by formula; gender counts by COUNTIF (424 / 93).
- **Visualise:** chart of married males vs females.
- **Interpret/Report:** findings recorded beneath the chart; saved as .xlsx.